

VENDOR PERFORMANCE ANALYSIS

Business-focused Data Analytics Report
Sales • Procurement • Pricing • Inventory • Profitability

\$67.99M	65.69%	\$2.71M
leading vendor sales	top-10 purchase contribution	estimated unsold inventory capital

Executive purpose

Evaluate vendor performance using commercial and operational measures, identify concentration and inventory risks, quantify purchasing opportunities, and test whether sales volume is aligned with profitability.

Data foundation: the project starts from a large ingestion database (approximately 2 GB and around 1 crore raw rows) and uses the consolidated vendor-sales summary for the analytical layer. The raw ingestion volume is therefore intentionally not presented as the number of records analysed in this report.

Prepared from the executed Vendor Performance Analysis notebook

1. Executive Summary

The analysis reviews vendor and product performance across sales, procurement, pricing, inventory movement and profitability. The objective is not simply to rank vendors, but to understand where the business is exposed to concentration risk, where purchasing economics can be improved, and where profitable products may be underperforming on sales.

\$67.99M	\$10.78	41.57%
DIAGEO total sales	large-order average unit price	low-sales group mean margin

Management takeaways

- **Supplier concentration is material:** the top 10 vendors account for 65.69% of total purchase dollars.
- **Volume creates a procurement advantage:** average unit purchase price falls from \$39.07 for small orders to \$10.78 for large orders.
- **Inventory needs attention:** approximately \$2.71M is estimated to be tied up in unsold inventory.
- **Sales and margin are not aligned:** the low-sales group has a higher mean profit margin than the high-sales group.
- **High-margin, low-sales products are an opportunity:** the issue may be reach, distribution or positioning rather than basic profitability.

Senior-analyst view: vendor quality should not be defined by sales volume alone. A practical vendor scorecard should combine sales, gross profit, margin, stock turnover and procurement contribution.

2. Data, Scope and Quality Assessment

The project extracts a consolidated **vendor_sales_summary** table from a SQLite inventory database and analyses it with Python/Pandas. The table combines vendor, brand and product attributes with purchase price, actual price, quantities, purchase dollars, sales dollars, tax, freight, gross profit, profit margin, stock turnover and sales-to-purchase ratio.

Data-quality observations

- PurchasePrice has a maximum of **\$5,681.81** versus a mean of approximately \$24.39, indicating a strongly skewed distribution.
- ActualPrice reaches **\$7,499.99** versus a mean of approximately \$35.64, again showing substantial skew.
- GrossProfit contains negative records, with a minimum of approximately **-\$52,002.78**.
- ProfitMargin contains $-\infty$ values, indicating undefined/non-finite ratio calculations for some records.
- TotalSalesQuantity and TotalSalesDollars include zero values, indicating records with purchasing activity but no recorded sales.
- StockTurnover ranges from **0 to 274.5**, showing very different inventory movement patterns.
- FreightCost varies widely, reaching approximately **\$257,032** at the upper end.

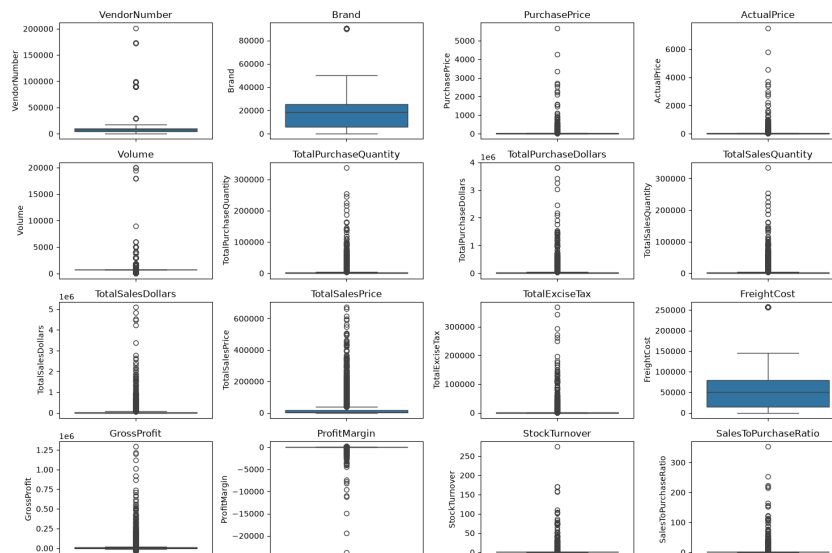


Figure 1. Boxplots used in the notebook to assess distributions and extreme values.

Reporting implication: these issues do not invalidate the business analysis, but they matter when building production dashboards. Non-finite margins, zero denominators and extreme values should be handled explicitly before automated KPI reporting.

3. Relationship Between Key Metrics

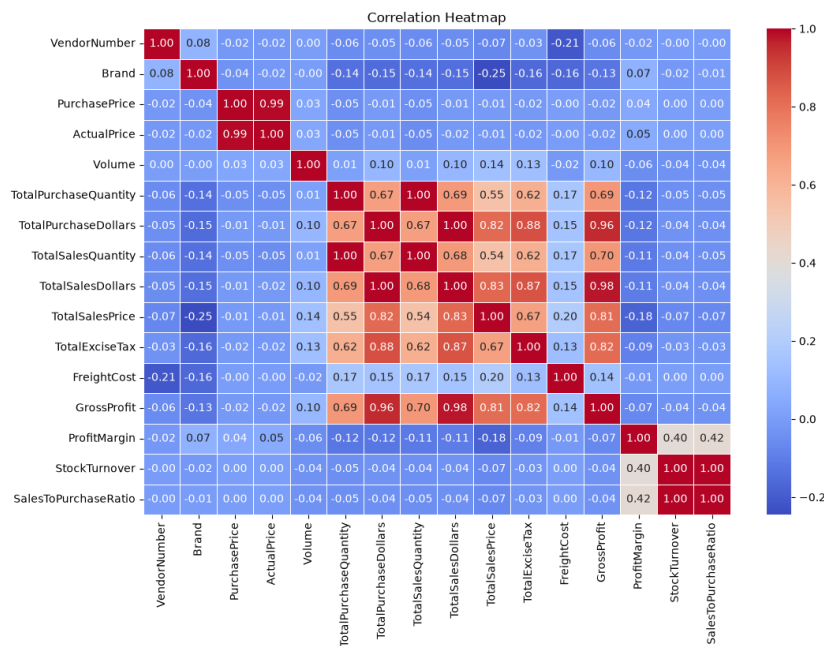


Figure 2. Correlation heatmap generated from the project dataset.

What the correlations tell us

- **PurchasePrice vs TotalSalesDollars: -0.012.** Unit purchase price alone has almost no linear relationship with sales revenue.
- **PurchasePrice vs GrossProfit: -0.016.** Profitability is influenced by more than purchase price.
- **TotalPurchaseQuantity vs TotalSalesQuantity: 0.999.** The two quantities move almost perfectly together in the analysed summary.
- **TotalSalesDollars vs GrossProfit: approximately 0.96.** Sales dollars and gross profit show a strong positive relationship.
- **ProfitMargin vs TotalSalesPrice: -0.179.** Higher sales-price values do not automatically imply higher margins.

Analytical caution: correlation is not causation. These relationships identify where deeper business investigation is warranted; they do not by themselves prove a causal relationship.

4. Sales Performance: Vendors and Products

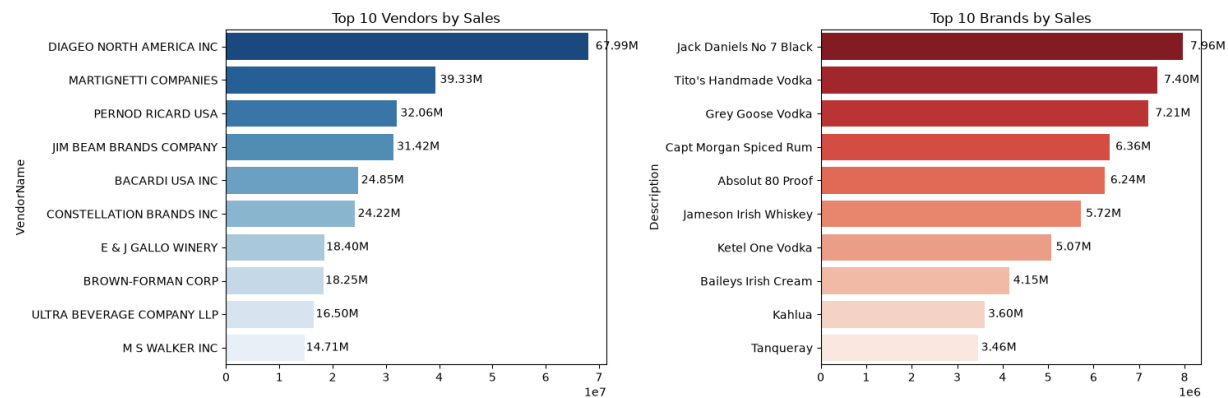


Figure 3. Top vendors and products by TotalSalesDollars.

DIAGEO NORTH AMERICA INC is the clear vendor leader at approximately **\$67.99M** in total sales. **MARTIGNETTI COMPANIES** follows at approximately \$39.33M, while **PERNOD RICARD USA** is around \$32.06M. **JIM BEAM BRANDS COMPANY** is also above \$31M.

Top products by sales

Rank	Product	Total Sales
1	Jack Daniels No 7 Black	\$7.96M
2	Tito's Handmade Vodka	\$7.40M
3	Grey Goose Vodka	\$7.21M
4	Capt Morgan Spiced Rum	\$6.36M
5	Absolut 80 Proof	\$6.24M
6	Jameson Irish Whiskey	\$5.72M
7	Ketel One Vodka	\$5.07M
8	Baileys Irish Cream	\$4.15M
9	Kahlua	\$3.60M
10	Tanqueray	\$3.46M

These rankings identify the vendor relationships and products with the greatest commercial exposure. They should receive closer monitoring when decisions are made around pricing, supply continuity and inventory availability.

5. Procurement Concentration & Supplier Risk

The top 10 vendors contribute **65.69%** of total purchase dollars. This concentration creates negotiating leverage, but also operational exposure if a major supplier experiences disruption.

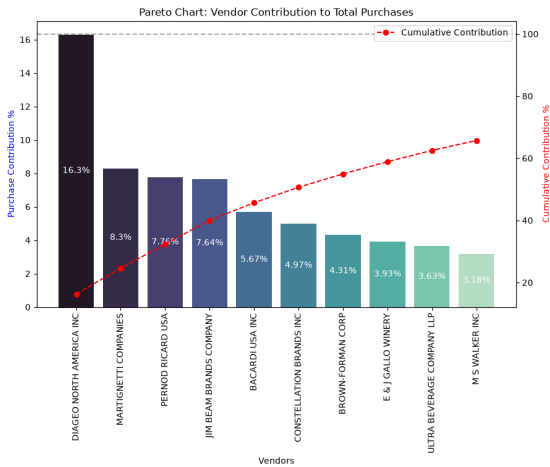


Figure 4. Pareto-style view of vendor contribution to total purchase dollars.

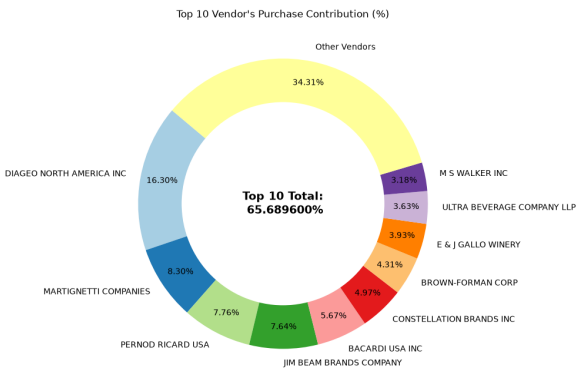


Figure 5. Top-vendor purchase contribution versus other vendors.

DIAGEO NORTH AMERICA INC contributes approximately 16.30% of total procurement, followed by **MARTIGNETTI COMPANIES** at 8.30% and **PERNOD RICARD USA** at 7.76%.

Management action: identify strategically important categories where a major supplier has no practical substitute. For those categories, maintain contingency suppliers and periodically review contract and service risk.

6. Purchasing Economics & Inventory Efficiency

Bulk purchasing and unit cost

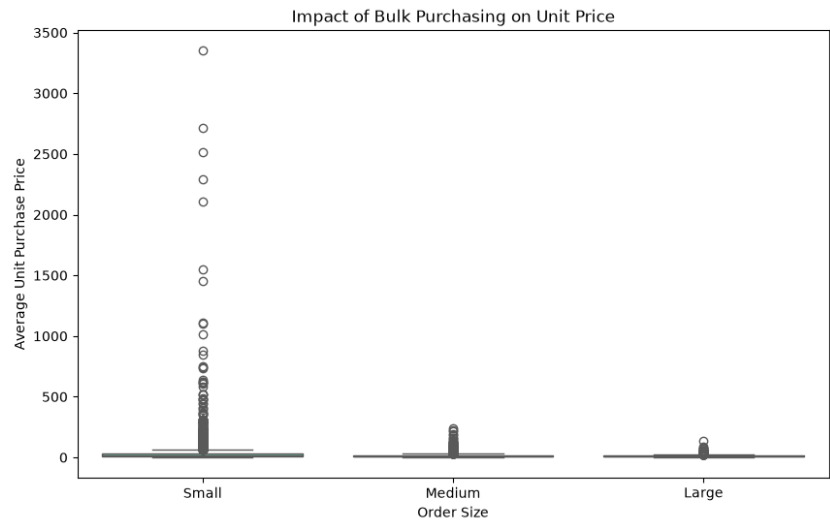


Figure 6. Average unit purchase price by order-size category.

Order size	Average unit purchase price
Small	\$39.07
Medium	\$15.49
Large	\$10.78

The large-order average unit price is approximately **72% lower** than the small-order average. This is a strong volume-pricing effect, but the saving is valuable only when additional inventory can be sold efficiently.

Inventory efficiency and capital at risk

The notebook flags vendors with average StockTurnover below 1 as potential slow-moving inventory cases. Examples include ALISA CARR BEVERAGES (0.615), HIGHLAND WINE MERCHANTS LLC (0.708) and PARK STREET IMPORTS LLC (0.751).

Vendor	Unsold inventory value
DIAGEO NORTH AMERICA INC	\$722.21K
JIM BEAM BRANDS COMPANY	\$554.67K
PERNOD RICARD USA	\$470.63K
WILLIAM GRANT & SONS INC	\$401.96K
E & J GALLO WINERY	\$228.28K

Total estimated unsold inventory capital is approximately \$2.71M. This is a working-capital opportunity: improve replenishment discipline, adjust purchase quantities for slow-moving items and use targeted promotions or assortment changes where appropriate.

7. Profitability: Statistical Validation

Vendors are split into high-sales and low-sales groups using the upper and lower quartiles of TotalSalesDollars. ProfitMargin is then compared using 95% confidence intervals and a Welch two-sample t-test.

Vendor group	Mean profit margin	95% confidence interval
High-sales group	31.18%	30.74% – 31.61%
Low-sales group	41.57%	40.50% – 42.64%

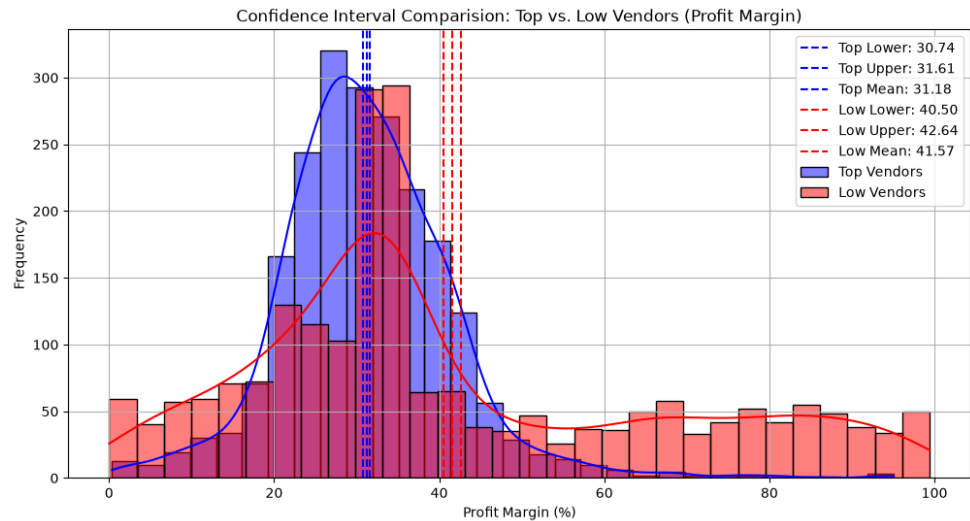


Figure 7. Profit-margin comparison and confidence intervals.

Hypotheses

H_0 : no significant difference in mean profit margins between the high-sales and low-sales groups.

H_a : a significant difference exists in mean profit margins between the two groups.

The executed Welch two-sample t-test reports approximately $t = -17.6695$ with $p = 0.0000$ as displayed in the notebook. At the 5% significance level, the null hypothesis is rejected.

Business interpretation: the low-sales group has the higher observed mean margin. This does not mean low sales are desirable; it means sales growth and margin optimization should be managed as separate objectives. The difference may reflect product mix, pricing or cost structure.

8. High-Margin / Low-Sales Growth Opportunity

The notebook identifies brands using a low-sales threshold of approximately **\$560.30** and a high-margin threshold of approximately **64.97%**, producing **198 target records**.

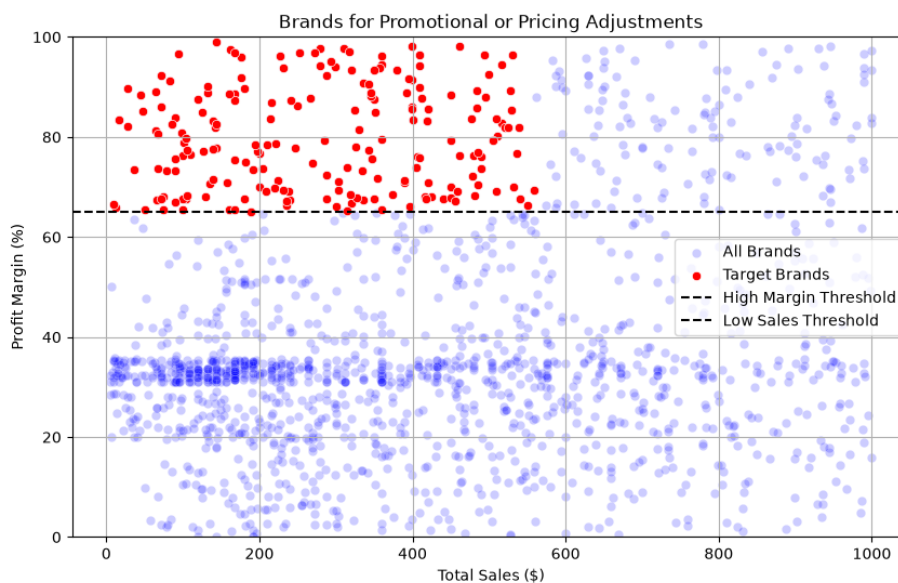


Figure 8. Brands identified as low-sales but high-margin opportunities.

These products already demonstrate attractive margins. The opportunity is therefore to investigate market reach, distribution, positioning and pricing rather than simply cutting price.

Recommended commercial response

- Test selective pricing or promotion rather than applying blanket discounts.
- Improve distribution and product availability in relevant channels.
- Use bundles or targeted campaigns where they can increase volume without destroying margin.
- Track incremental sales and margin after each intervention rather than judging success on revenue alone.

Decision principle: protect the existing margin advantage while solving the demand or reach constraint.

9. Final Recommendations

The recommendations below translate the analytical findings into practical management actions and retain the direction of the final recommendation section in the project.

1. Re-evaluate pricing for low-sales, high-margin brands

Re-evaluate pricing for low-sales, high-margin brands to drive higher sales volume without sacrificing profitability.

2. Diversify vendor partnerships

Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply-chain risks.

3. Leverage bulk purchasing advantages

Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.

4. Optimize slow-moving inventory

Optimize slow-moving inventory by adjusting purchase quantities, replenishment decisions and product-level promotions.

5. Enhance marketing and distribution

Enhance marketing and distribution strategies for low-performing vendors to drive higher sales volumes without compromising profit margins.

6. Manage sales and profitability together

Use a vendor scorecard combining sales, gross profit, profit margin, stock turnover and procurement contribution. A vendor should not be labelled good or bad from sales volume alone.

Expected business outcome: By implementing these recommendations, the company can achieve sustainable profitability, mitigate risks, and enhance overall operational efficiency.

10. Conclusion

The Vendor Performance Analysis moves from raw vendor-level data to business decisions. The analysis identifies the strongest vendors and products, quantifies procurement concentration, demonstrates a substantial volume-pricing effect, highlights slow-moving inventory and estimates the capital tied up in unsold stock.

The profitability section adds an important layer of validation. The statistically significant difference between the high-sales and low-sales groups shows that higher sales do not automatically translate into higher margins. Growth, margin and inventory efficiency therefore need to be considered together.

65.69%	~72%	41.57%
top-10 procurement share	small-to-large unit-price reduction	low-sales group mean margin

Overall assessment

From a data-analytics perspective, the project demonstrates a complete workflow: SQLite data extraction, Pandas-based analysis, exploratory visualization, business segmentation and statistical testing. The strongest part of the work is the connection between measurable patterns and practical decisions around procurement, pricing, inventory and sales.

Final note: the figures and interpretations in this report are based on the executed notebook results. Raw-data issues such as non-finite profit-margin values are reported as findings rather than silently removed. Values shown in the report are rounded for readability, without changing the underlying conclusions.